AUH Twin

Can aviation data estimate international hotel demand in Abu Dhabi?
AUH Twin follows seats on every route to hotel guests, month by month, and shows how a flight change would move them.

106 DEPARTURE CITIES | **33** COUNTRIES | **45** NATIONALITIES

FROM SEATS TO STAYS, ONE YEAR

Aug 2025 to Jul 2026. Hotel share and days are model estimates.

14.6M Seats — on the flight schedule

85% full

12.5M Passengers — seats that are filled

49% stay

6.1M Visitors — end their trip here

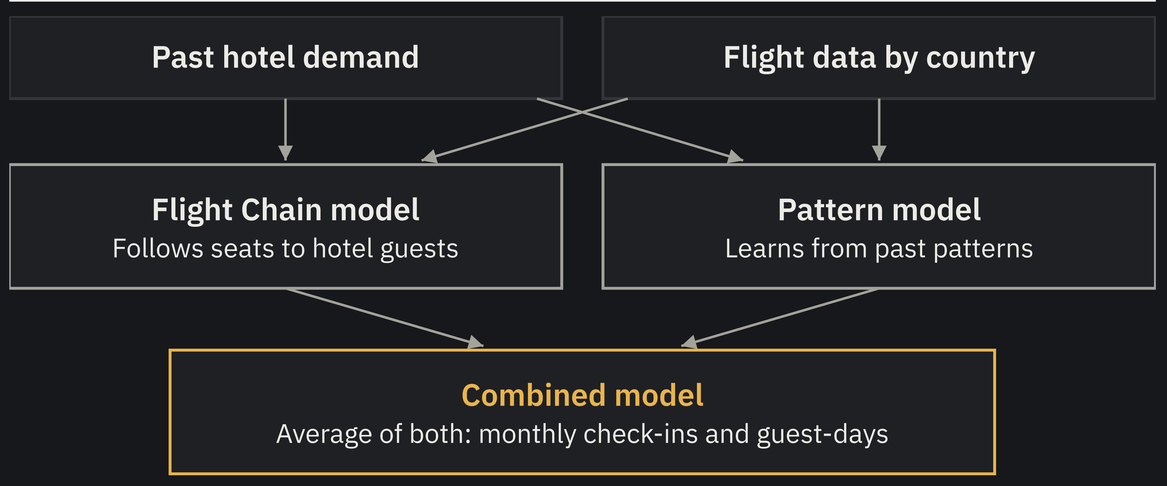
46% in hotels

2.79M Hotel guests — check into a hotel

3.5 days each

9.86M Guest-days — by nationality and month

FIG. 1 HOW IT WORKS



Error: average gap to the real figure. Guest-day: one guest in a hotel for one day.

HOW ACCURATE IT IS (AVERAGE ERROR)

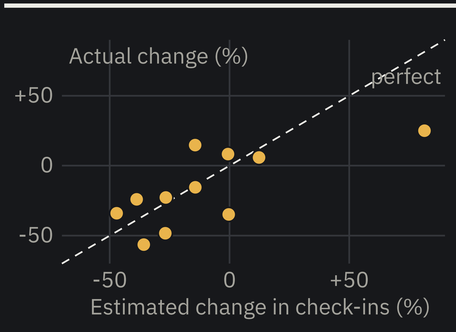
16.5% next month's guest-days, forecast in advance. Using last year's figure gives 23.7%.

15.5% next month, with real flight numbers. Likely range 14.1 to 17.0%.

18.4% by nationality, 1 to 7 months ahead. Better than last year's figure (26.7%) in all 23 tests.

7.3% on the monthly total, 1 to 7 months ahead. Last year's figure gives 21.3%.

FIG. 2 DEMAND REACTS TO REAL AIRLINE CHANGES

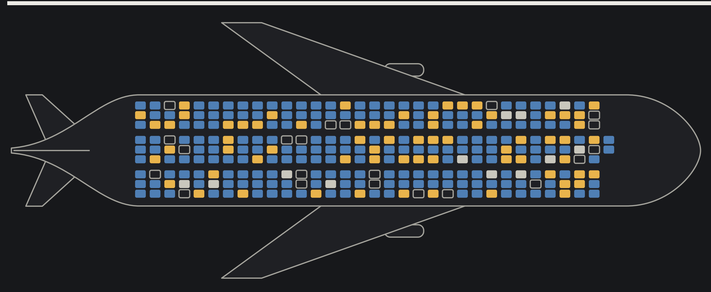


Each dot is an airline starting or stopping a route, 11 cases.

Dots near the dashed line: the estimate was right.

Real demand moved about **0.6 to 0.8** as much as estimated.

FIG. 3 ONE MORE LONDON FLIGHT, SEAT BY SEAT



Check into a hotel	67
Visit, not in a hotel	12
Connect onward	190
Empty seat	21

67 guests × 4.7 days × 365 flights
= about 115,000 guest-days

290 seats. Hotel share and stay are model estimates.

FIG. 4 THE COMBINED MODEL IS BEST OR EQUAL ON EVERY MEASURE

	Guest-days	Check-ins
Using last year's figure	23.7%	24.0%
Flights only	20.5%	19.8%
Simple growth formula	18.4%	19.5%
Flight Chain model	17.1% / 16.3%	17.1% / 16.3%
Pattern model	16.9% / 15.5%	18.1% / 16.3%
Combined model	16.5% / 15.5%	16.9% / 15.8%

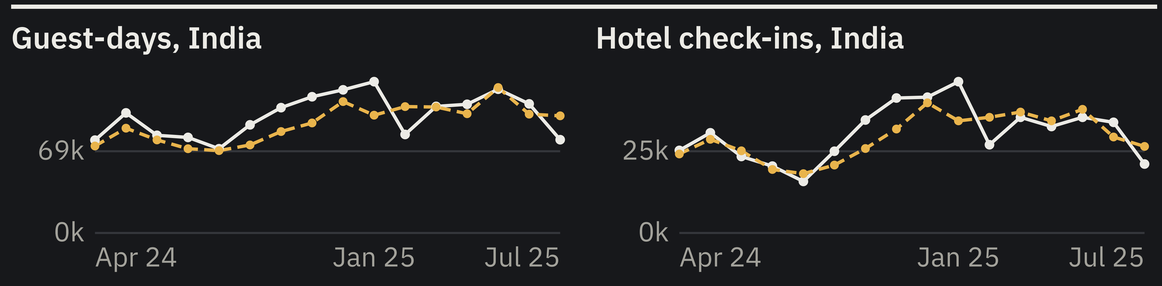
In advance | With real flight numbers

Next-month error, Apr 2024 to Jul 2025. In advance: known before the month starts. Real flight numbers: known after it is flown.

FIG. 5 WHAT WE LEARNED

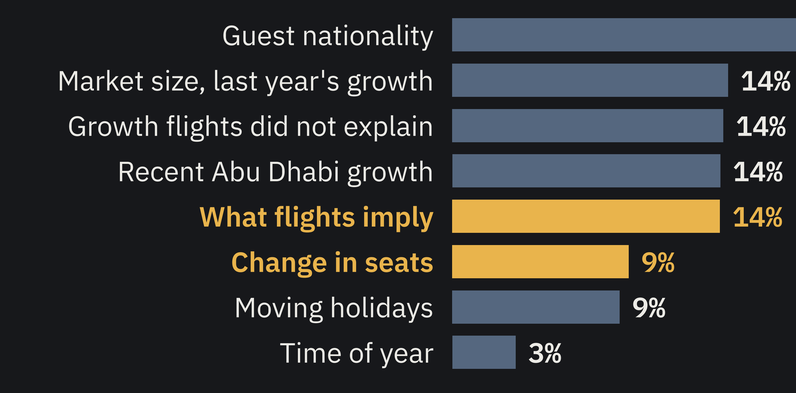
- Two different models, combined, are as good as or better than either alone.
- A simple growth formula: 18.4% error. The combined model: 16.5%.
- Real flight numbers make both models more accurate on guest-days.
- Guest-days: the combined model beat last year in 15 of 16 months.

FIG. 6 ESTIMATES FOLLOW REAL DEMAND



Solid: real figures. Dashed: Flight Chain model. India is the largest market.

FIG. 7 WHAT DRIVES THE MACHINE-LEARNING CORRECTION



Gold: flight signals. Covers the Flight Chain machine learning only. Influence, not cause.

IMPACT FOR PLANNERS

+115,000
guest-days a year, one more daily London flight

+507,000
guest-days a year, best 5,000 extra weekly seats

2.4M
guest-days a year at risk if all India flights stop

HOW IT WAS TESTED

Tested 23 times
- each time using only earlier months

No peeking at answers
- later data had no effect

Rebuilt independently
- same scores for both models

FROM PROTOTYPE TO PILOT

Done
- [x] What-if tool running
- [x] Two models combined
- [x] Drivers explained
- [x] Airline changes checked

Next
- [ ] Live monthly forecasts
- [ ] Add forecast ranges
- [ ] Monthly retraining
- [ ] Pilot with planners

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